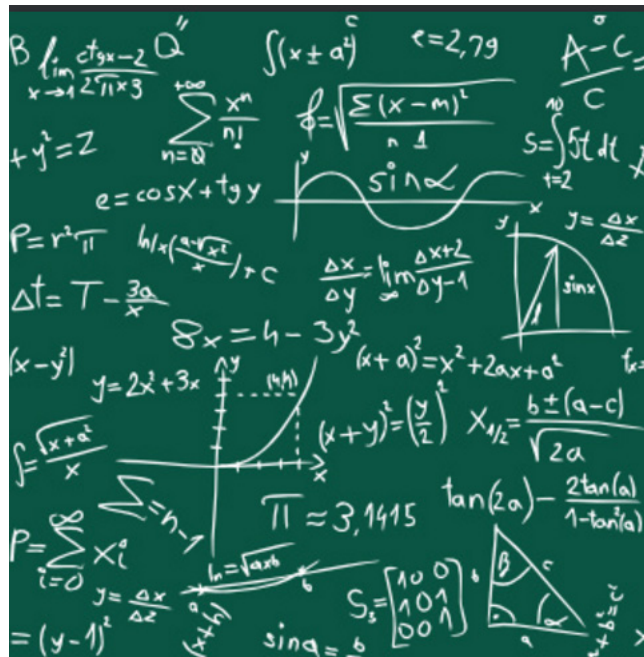


Mean value theorem, maxima and minima



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani e Sandro Salsa, Zanichelli.

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1 Local and global maxima/minima

- One of the uses of differential calculus is the **search for maxima and minima**, that is, the **optimization** of a function defined on an interval I (closed/open, bounded/unbounded).

Definition 1: of maximum point and maximum

Given a function $f : I \rightarrow \mathbb{R}$, if there exists a point $\tilde{x}_M \in I$ such that:

$$f(\tilde{x}_M) \geq f(x), \quad \forall x \in I$$

then $\tilde{x}_M$ is a (global) **maximum point** and $f(\tilde{x}_M)$ is the (global) **maximum** of f in I .

Definition 2: of minimum point and minimum

Given a function $f : I \rightarrow \mathbb{R}$, if there exists a point $\tilde{x}_m \in I$ such that:

$$f(\tilde{x}_m) \leq f(x), \quad \forall x \in I$$

then $\tilde{x}_m$ is a (global) **minimum point** and $f(\tilde{x}_m)$ is the (global) **minimum** of f in I .

We call an **extremum** a maximum or a minimum, and an **extremum point** a maximum or minimum point. An extremum, if it exists, is unique, while there may be several extremum points (even infinitely many).

Definition 3: of local maximum point and local maximum

Given a function $f : I \rightarrow \mathbb{R}$, if there exist a point $\bar{x}_M \in I$ and a neighborhood $(\bar{x}_M - \delta, \bar{x}_M + \delta)$ with $\delta > 0$, such that:

$$f(\bar{x}_M) \geq f(x), \quad \forall x \in (\bar{x}_M - \delta, \bar{x}_M + \delta) \cap I$$

then $\bar{x}_M$ is a **local maximum point** and $f(\bar{x}_M)$ is the **local maximum** of f in $(\bar{x}_M - \delta, \bar{x}_M + \delta) \cap I$.

Definition 4: of local minimum point and local minimum

Given a function $f : I \rightarrow \mathbb{R}$, if there exist a point $\bar{x}_m \in I$ and a neighborhood $(\bar{x}_m - \delta, \bar{x}_m + \delta)$ with $\delta > 0$, such that:

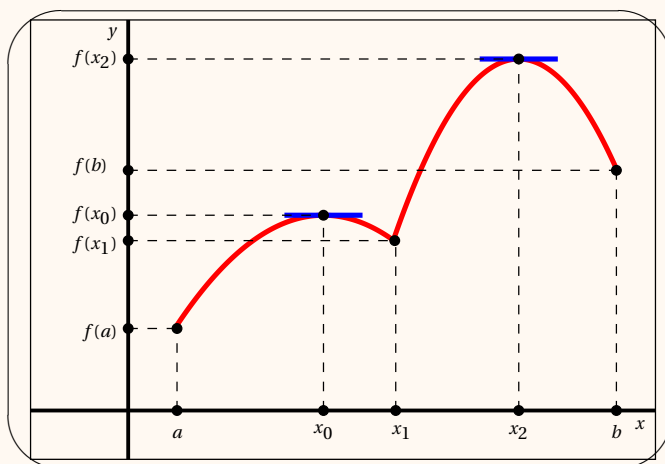
$$f(\bar{x}_m) \leq f(x), \quad \forall x \in (\bar{x}_m - \delta, \bar{x}_m + \delta) \cap I$$

then $\bar{x}_m$ is a **local minimum point** and $f(\bar{x}_m)$ is the **local minimum** of f in $(\bar{x}_m - \delta, \bar{x}_m + \delta) \cap I$.

- Global extremum points are also local extremum points, and global extrema are also local extrema.

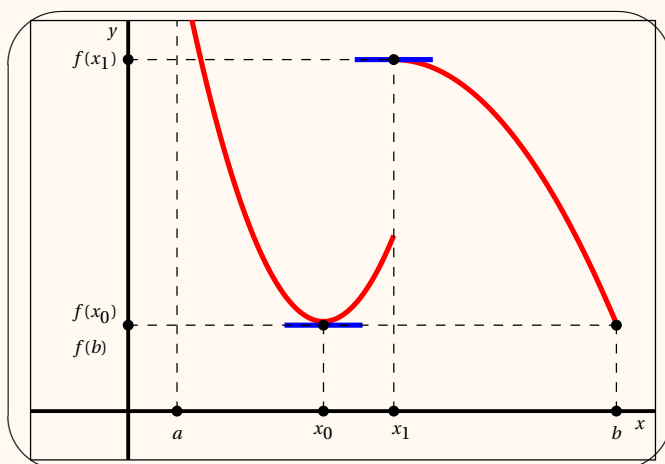
Example 1: extrema and extremum points (global and local)

Consider the function f with the following graph:



- the global maximum is $f(x_2)$ and x_2 is a global maximum point (it is also unique); $f(x_0)$ is a local (not global) maximum and x_0 is a local (not global) maximum point
- the global minimum is $f(a)$ and a is a global minimum point (it is also unique); $f(b)$ and $f(x_1)$ are two local (not global) minima and b and x_1 are two local (not global) minimum points

Now consider the function f with the following graph:



- the global maximum does not exist ($\lim_{x \rightarrow a^+} f(x) = +\infty$); $f(x_1)$ is a local (not global) maximum and x_1 is a local (not global) maximum point
- the global minimum is $f(x_0)$, which is also equal to $f(b)$; x_0 and b are two global minimum points

2 Fermat's theorem and stationary points

- At a local or global extremum point the function may fail to be differentiable and may even be discontinuous. However, the following theorem tells us that if a function is differentiable at a local extremum point, then at that point the derivative vanishes and hence the tangent to the graph is horizontal.

Theorem 1: Fermat's theorem

Given a function $f : (a, b) \rightarrow \mathbb{R}$ differentiable at $x_0 \in (a, b)$, if f has a local extremum at x_0 then $f'(x_0) = 0$.

Proof. Consider the case where x_0 is a local maximum point. Then:

$$\exists(x_0 - \delta, x_0 + \delta) \text{ with } \delta > 0: f(x_0) \geq f(x), \forall x \in (x_0 - \delta, x_0 + \delta) \cap (a, b)$$

Therefore for every $x \in (x_0 - \delta, x_0 + \delta) \cap (a, b)$ we have:

$$x < x_0 \Rightarrow \frac{f(x) - f(x_0)}{x - x_0} \geq 0 \text{ and hence } f'_-(x_0) = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} \geq 0$$

where for the non-negativity of the limit we used the sign-preservation theorem. On the other hand:

$$x > x_0 \Rightarrow \frac{f(x) - f(x_0)}{x - x_0} \leq 0 \text{ and hence } f'_+(x_0) = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0} \leq 0$$

Since f is differentiable at x_0 , we have:

$$f'(x_0) = f'_-(x_0) = f'_+(x_0) = 0$$

The case where x_0 is a local minimum point is handled in the same way. $\square$

Proof. Suppose, by contradiction, that $f'(x_0) > 0$; then by the sign-preservation theorem we would have $\frac{f(x) - f(x_0)}{x - x_0} > 0$ eventually as $x \rightarrow x_0$. Taking into account the sign of $x - x_0$, this implies that $f(x) > f(x_0)$ eventually as $x \rightarrow x_0^+$ and $f(x) < f(x_0)$ eventually as $x \rightarrow x_0^-$. But this contradicts the hypothesis that x_0 is a local extremum point of f . Similarly, we rule out the case $f'(x_0) < 0$. Therefore we must have $f'(x_0) = 0$. $\square$

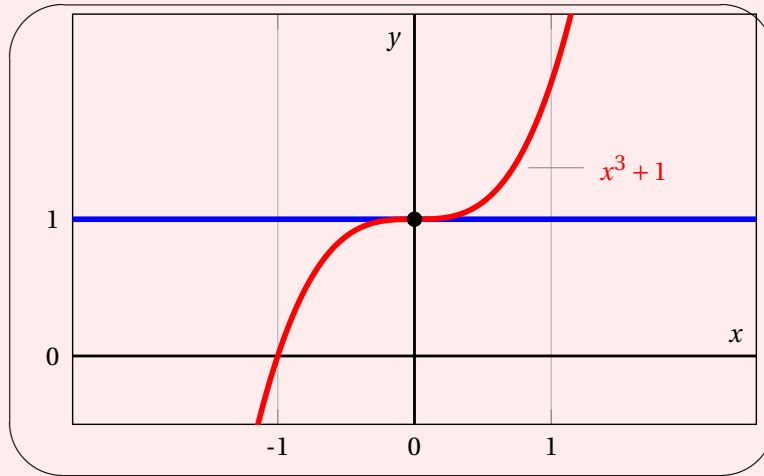
Definition 5: stationary point

A point x_0 is called a stationary point of f if f is differentiable at x_0 and $f'(x_0) = 0$.

Fermat's theorem says that for a function $f : (a, b) \rightarrow \mathbb{R}$ differentiable at $x_0 \in (a, b)$ we have:

$$x_0 \text{ is a local extremum point} \Rightarrow \underbrace{x_0 \text{ is a stationary point}}_{f'(x_0)=0} \quad (1)$$

Hence “ x_0 is a local extremum point” is a sufficient condition for “ x_0 is a stationary point”. But the converse is not true. A counterexample is given by the function $f(x) = x^3 + 1, \forall x \in \mathbb{R}$, whose derivative function is $f'(x) = 3x^2, \forall x \in \mathbb{R}$. With $x_0 = 0$ we have $f'(0) = 0$ but $x_0 = 0$ is not a local extremum point.



Hence we have:

$$\underbrace{x_0 \text{ is a stationary point}}_{f'(x_0)=0} \not\Rightarrow x_0 \text{ is a local extremum point}$$

Finally, from the contrapositive of (1), we have:

$$x_0 \text{ is not a stationary point} \Rightarrow x_0 \text{ is not a local extremum point}$$

3 Mean value theorem (Lagrange's theorem)

Theorem 2: mean value theorem (Lagrange)

Given a function f differentiable in (a, b) and continuous in $[a, b]$, then:

$$\exists c \in (a, b) : \frac{f(b) - f(a)}{b - a} = f'(c) \quad (2)$$

In the case $f(b) = f(a)$, the theorem tells us that there exists $c \in (a, b)$ such that $f'(c) = 0$, that is, there exists at least one point where the derivative vanishes. This corollary of Lagrange's theorem is called **Rolle's theorem**.

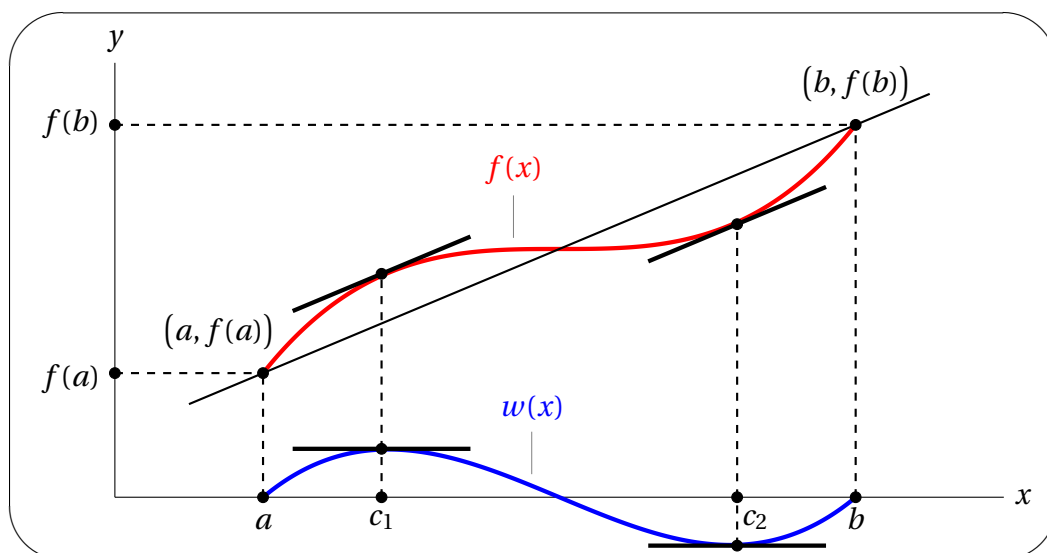
- Differentiability is required only on the open interval (a, b) in order to include also functions that are continuous at one of the endpoints but not differentiable there. For example, with $a = 0$:

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) x & \text{if } x \in (0, b) \\ 0 & \text{if } x = 0 \end{cases} \quad \text{or} \quad g(x) = \sqrt{x}$$

- Geometrically, considering the graph of f we have:

1. $\frac{f(b) - f(a)}{b - a}$ is the slope of the line through $(a, f(a))$ and $(b, f(b))$
2. $f'(c)$ is the slope of the tangent line to the graph of f at the point $(c, f(c))$

The mean value theorem therefore expresses the fact that at the point $(c, f(c))$ the tangent to the graph of f is parallel to the line through $(a, f(a))$ and $(b, f(b))$. There may be more than one point with this property, as the following figure shows:



- The figure also shows the function

$$w(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right)$$

given by the difference between the values of the function and the line through the points $(a, f(a))$ and $(b, f(b))$, which has equation:

$$y = f(a) + \frac{f(b) - f(a)}{b - a} (x - a)$$

The function $w(x)$ is the key to the proof of the theorem.

Proof. Consider the function:

$$w(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right)$$

Clearly $w(a) = w(b) = 0$; moreover w is continuous in $[a, b]$ and differentiable in (a, b) . Since

$$w'(x) = f'(x) - \frac{f(b) - f(a)}{b - a} \quad \text{then} \quad \frac{f(b) - f(a)}{b - a} = f'(c) \iff \exists c \in (a, b) : w'(c) = 0$$

Since w is continuous in $[a, b]$, by the Weierstrass theorem there exist two points x_1 and x_2 in $[a, b]$ such that:

$w(x_1) = M$, the maximum of w in $[a, b]$; $w(x_2) = m$, the minimum of w in $[a, b]$.

If $M = m$, then $w(x)$ is constant in $[a, b]$, and hence $w'(x) = 0, \forall x \in (a, b)$.

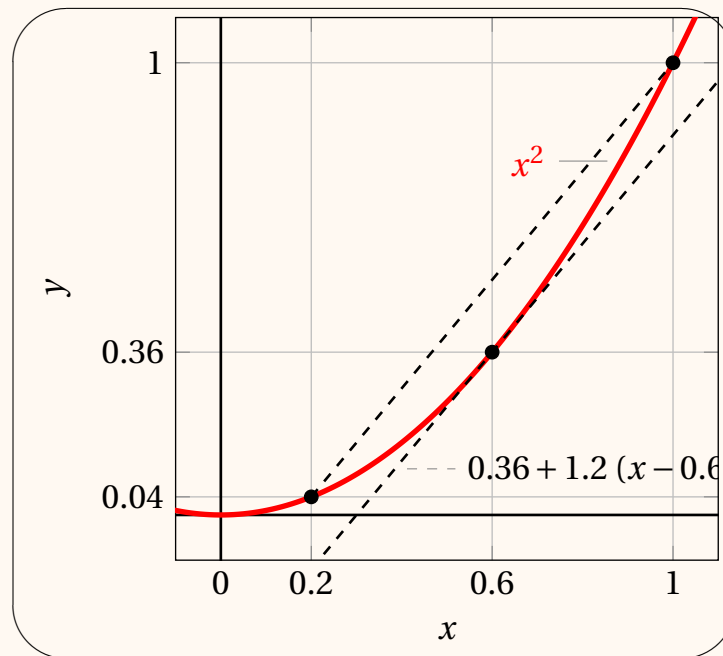
If $M > m$, at least one of the two points x_1 or x_2 is not at the endpoints of the interval, since $w(a) = w(b) = 0$. Fermat's theorem then implies that at the maximum or minimum point that lies in the interior (possibly both) the derivative of w vanishes, and the theorem is thus proved. $\square$

Example 2: using the mean value theorem

Let $f(x) = x^2$. Then $f'(x) = 2x$ and the mean value theorem states that in every interval $[a, b]$ there exists a number c such that:

$$\frac{b^2 - a^2}{b - a} = 2c \quad \text{from which} \quad c = \frac{a + b}{2} \quad (\text{arithmetic mean})$$

That is, every chord AB of the parabola $y = x^2$ is parallel to the tangent at the point whose abscissa equals the arithmetic mean of the abscissas of A and B . Taking for example the interval $[0.2, 1]$ ($a = 0.2, b = 1$ and $c = 0.6$), we have:

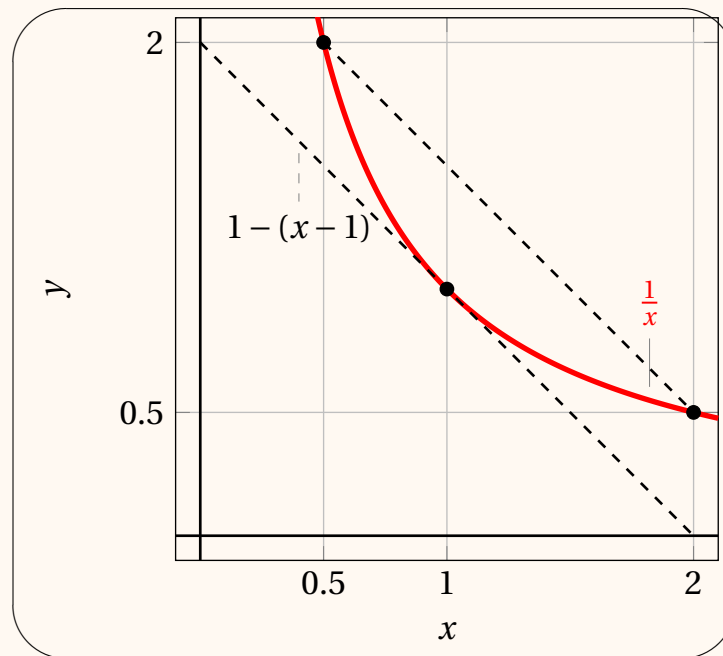


Example 3: using the mean value theorem

Let $f(x) = \frac{1}{x}$. Then $f'(x) = -\frac{1}{x^2}$ and the mean value theorem states that in every interval $[a, b]$ there exists a number c such that:

$$\frac{\frac{1}{b} - \frac{1}{a}}{b - a} = -\frac{1}{c^2} \quad \text{from which} \quad c = \sqrt{a \cdot b} \quad (\text{geometric mean})$$

That is, every chord AB of the hyperbola $y = \frac{1}{x}$ is parallel to the tangent at the point whose abscissa equals the geometric mean of the abscissas of A and B . Taking for example the interval $[0.5, 2]$ ($a = 0.5, b = 2$ and $c = 1$), we have:



4 Differential monotonicity test theorem

Theorem 3: Differential Monotonicity Test

Given a function $f : I \rightarrow \mathbb{R}$, continuous in I and differentiable at the interior points of I , then:

$$f'(x) \geq 0, \forall x \text{ in the interior of } I \iff f \text{ is non-decreasing in } I \quad (3)$$

$$f'(x) \leq 0, \forall x \text{ in the interior of } I \iff f \text{ is non-increasing in } I \quad (4)$$

$$f'(x) > 0, \forall x \text{ in the interior of } I \implies f \text{ is increasing in } I \quad (5)$$

$$f'(x) < 0, \forall x \text{ in the interior of } I \implies f \text{ is decreasing in } I \quad (6)$$

Proof.

($\Rightarrow$) Taking two points $x_1 < x_2$ in the interval I , we can apply Lagrange's theorem on the interval $[x_1, x_2]$, hence there exists $c \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

If $f'(c) \geq 0$ (resp. > 0), then $f(x_2) \geq f(x_1)$ (resp. $f(x_2) > f(x_1)$). Since x_1 and x_2 are arbitrary, we have proved that f is non-decreasing (resp. increasing) in I . If $f'(c) \leq 0$ (resp. < 0) the argument is analogous.

($\Leftarrow$) If f is non-decreasing, then we have

$$\frac{f(z) - f(x)}{z - x} \geq 0 \quad \text{for all } z, x \in I, z \neq x$$

and, by the sign-preservation theorem, the limit of the ratio as $z \rightarrow x$ is also non-negative; by hypothesis this limit exists and equals $f'(x)$ for every x in the interior of I . If f is non-increasing the argument is analogous.

□

The Differential Monotonicity Test theorem says that for a function $f : I \rightarrow \mathbb{R}$ differentiable at the interior points of I we have:

$$f'(x) \geq 0, \forall x \text{ in the interior of } I \iff f \text{ is non-decreasing in } I$$

Hence “ $f'(x) \geq 0, \forall x$ in the interior of I ” is a necessary and sufficient condition for “ f is non-decreasing in I ”. The same holds for the case of non-increasing functions.

The Differential Monotonicity Test theorem says that for a function $f : I \rightarrow \mathbb{R}$ differentiable at the interior points of I we have:

$$f'(x) > 0, \forall x \text{ in the interior of } I \implies f \text{ is increasing in } I$$

Hence “ $f'(x) > 0, \forall x$ in the interior of I ” is a sufficient condition for “ f is increasing in I ”. But the converse is not true. A counterexample is given by the function $f(x) = x^3 + 1, \forall x \in \mathbb{R}$, which is increasing and differentiable in $\mathbb{R}$, but $f'(x) = 3x^2$ vanishes at $x_0 = 0$.

Hence we have:

$$f \text{ is increasing in } I \not\Rightarrow f'(x) > 0, \forall x \text{ in the interior of } I$$

Finally, from the contrapositive of (5), we have:

$$f \text{ is not increasing in } I \Rightarrow \exists x \text{ in the interior of } I : f'(x) \leq 0,$$

To be checked. The same holds for the case of decreasing functions.

- It follows immediately from the theorem that with $I = (a, b)$ we have:

$$f'(x) = 0, \forall x \in (a, b) \implies f \text{ constant in } (a, b)$$

The converse implication is obvious. We then have

$$f'(x) = 0, \forall x \in (a, b) \iff f \text{ constant in } (a, b) \quad (7)$$

Hence “ $f'(x) = 0, \forall x \in (a, b)$ ” is a necessary and sufficient condition for “ f constant in (a, b) ”.

Example 4: functions with zero derivative

Consider the function:

$$f(x) = \arctan x + \arctan \frac{1}{x}, \quad \forall x \neq 0$$

we have

$$f'(x) = \frac{1}{1+x^2} + \frac{1}{1+\frac{1}{x^2}} \left(-\frac{1}{x^2}\right) = 0, \quad \forall x \neq 0$$

Implication (7) allows us to say that f is constant on the interval $(-\infty, 0)$ and on the interval $(0, \infty)$. To find its value, it is enough to compute f at one point of each interval, for example:

$$f(1) = \arctan(1) + \arctan(1) = 2 \frac{\pi}{4} = \frac{\pi}{2}$$

$$f(-1) = \arctan(-1) + \arctan(-1) = -\frac{\pi}{2}$$

We have therefore proved that:

$$\arctan x + \arctan \frac{1}{x} = \begin{cases} \frac{\pi}{2} & \text{if } x > 0 \\ -\frac{\pi}{2} & \text{if } x < 0 \end{cases}$$

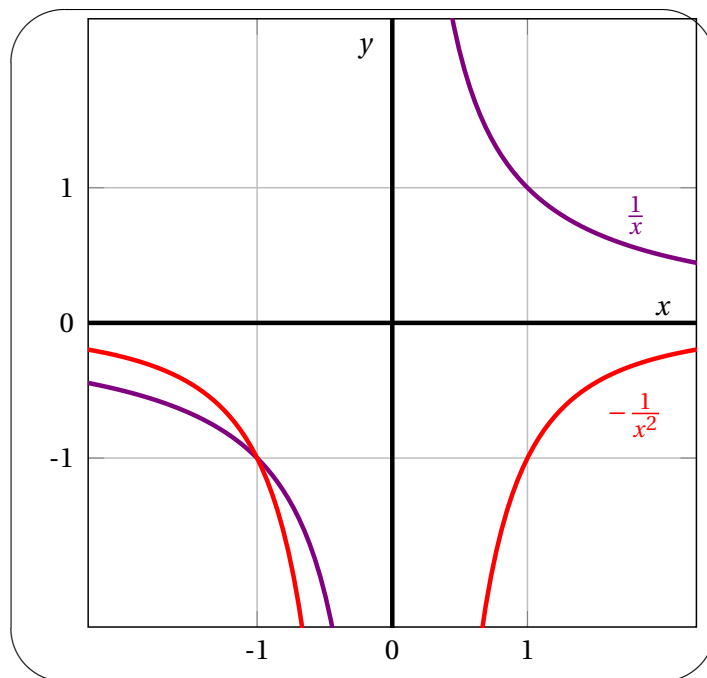
- Both in the Differential Monotonicity Test theorem and in implication (7) it is essential that the set I is an interval. For example, the function

$$f(x) = \frac{1}{x}, \quad \forall x \in \mathbb{R} \setminus \{0\}$$

has derivative

$$f'(x) = -\frac{1}{x^2} < 0, \quad \forall x \in \mathbb{R} \setminus \{0\}$$

but the function is not decreasing on its domain. The function is decreasing on the interval $(-\infty, 0)$ and on the interval $(0, \infty)$, as the figure shows:



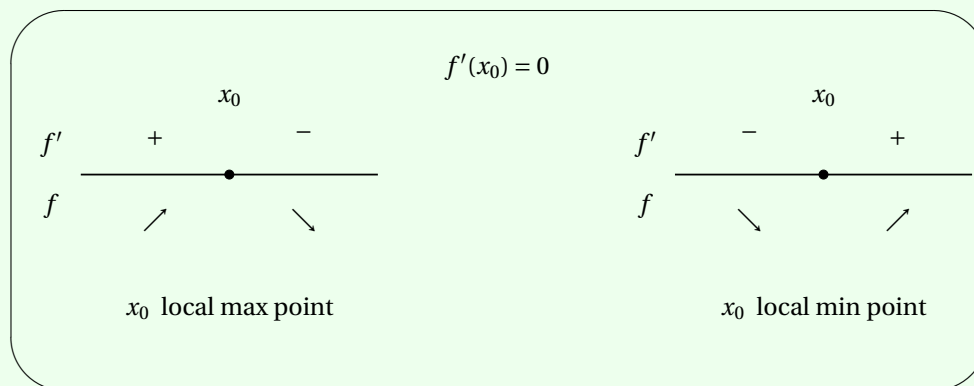
5 Finding maxima and minima

Suppose we have a function $f : [a, b] \rightarrow \mathbb{R}$ and we want to find its local and global maxima and minima in $[a, b]$. If f is differentiable in (a, b) we can proceed as follows:

Step 1 Compute $f(a)$ and $f(b)$.

Step 2 Compute $f'(x)$ and solve the equation $f'(x) = 0$. In this way we find the stationary points, among which are the possible local extremum points in the interval (a, b) .

Step 3 If there are no stationary points, $f(a)$ and $f(b)$ are the global extrema and a and b are global extremum points. Otherwise we need to determine the nature of the stationary points found. The function f has a local extremum at a stationary point if and only if the sign of f' changes there, which we check by studying the sign of f' in a neighborhood of the point. We have the following cases:



Step 4 Having found the possible local extremum points, compute the value of f at these points and compare it with $f(a)$ and $f(b)$ to understand whether or not they are global extremum points; in this way the global extrema are determined.

Example 5: finding maxima and minima

Consider the function:

$$f(x) = x e^{-x^2}, \text{ with } x \in [0, 2]$$

Step 1:

$$f(0) = 0, \quad f(2) = 2 e^{-4}$$

Example 6: finding maxima and minima

Step 2:

$$f'(x) = e^{-x^2} + x e^{-x^2} (-2x) = e^{-x^2} (1 - 2x^2)$$

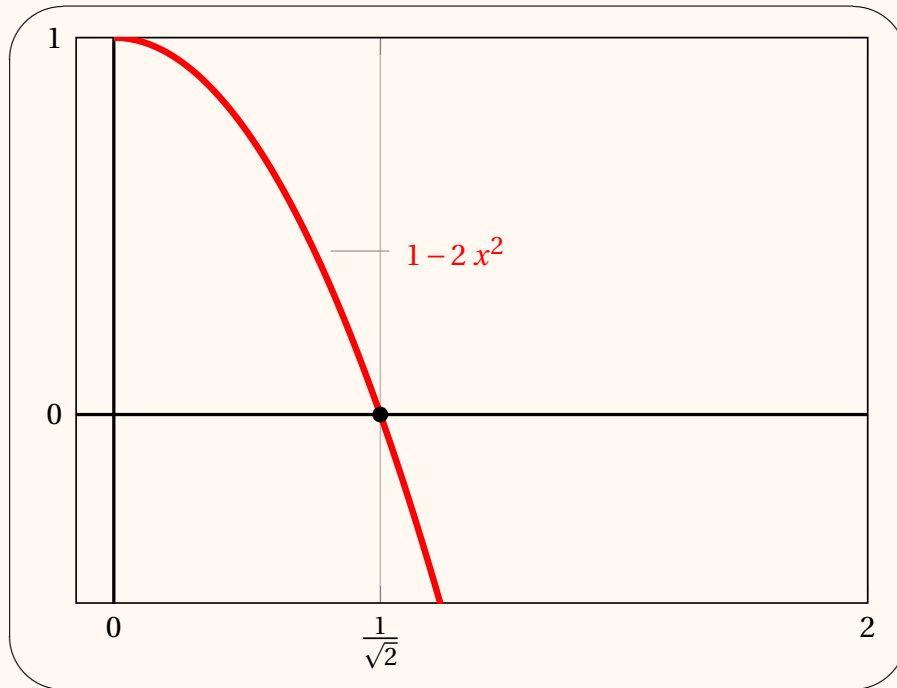
We have $e^{-x^2} > 0, \forall x \in \mathbb{R}$, hence:

$$f'(x) = 0 \iff (1 - 2x^2) = 0 \iff x = \pm \frac{1}{\sqrt{2}}$$

Only $\frac{1}{\sqrt{2}} \in [0, 2]$, hence we have the stationary point $x_0 = \frac{1}{\sqrt{2}}$.

Step 3:

We study the sign of f' near $x_0 = \frac{1}{\sqrt{2}}$. We have $f'(x) \geq 0$ for $2x^2 \leq 1$, that is, for $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$



We therefore conclude that:

$\frac{1}{\sqrt{2}}$ is a local maximum point and $f\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} e^{-1/2} = \frac{1}{\sqrt{2}e}$ is a local maximum

We also note that:

$$\lim_{x \rightarrow 0^+} \frac{x e^{-x^2}}{x} = \lim_{x \rightarrow 0^+} \frac{1}{e^{x^2}} = 1 \quad \text{hence} \quad f(x) \sim x \quad \text{for} \quad x \rightarrow 0^+$$

Example 7: finding maxima and minima

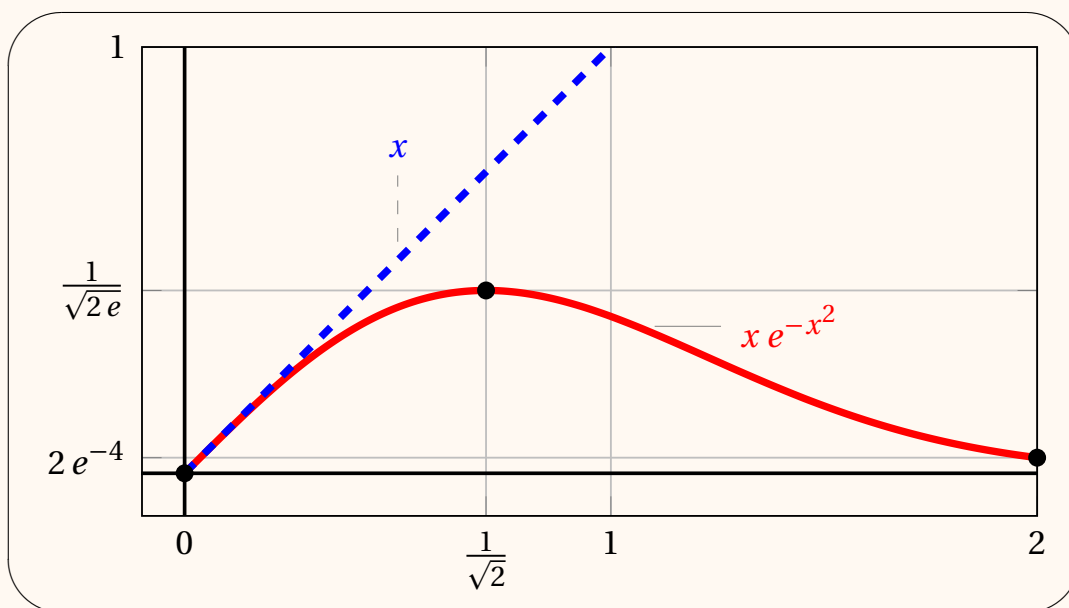
Step 4:

$$\frac{1}{\sqrt{2}e} > f(0) = 0, \quad \frac{1}{\sqrt{2}e} > f(2) = 2e^{-4}$$

We therefore conclude that:

$\frac{1}{\sqrt{2}}$ is the (unique) global maximum point and 0 is the (unique) global minimum point

$\frac{1}{\sqrt{2}e}$ is the global maximum and 0 is the global minimum



5.1 From the discrete to the continuous

Passing “from the discrete to the continuous” for sequences (i.e., from the natural numbers to the reals) is a way to make the tools of differential calculus available

Example 8: proving monotonicity of sequences by passing to the continuous

Consider the sequence

$$a_n = \frac{\log n}{n}, \quad \text{we have } a_n \geq 0, \forall n \in \mathbb{N}_{>0} \text{ and } a_n \rightarrow 0 \text{ for } n \rightarrow +\infty$$

To prove that the sequence is eventually monotone non-increasing, one way is to use the definition and prove that:

$$a_n \geq a_{n+1}, \quad \forall n \in \mathbb{N}_{>0} \quad \text{i.e.} \quad \frac{\log n}{n} \geq \frac{\log(n+1)}{n+1}, \quad \forall n \in \mathbb{N}_{>0}$$

Since both the numerator and the denominator grow as n grows, it is not easy to prove this inequality algebraically.

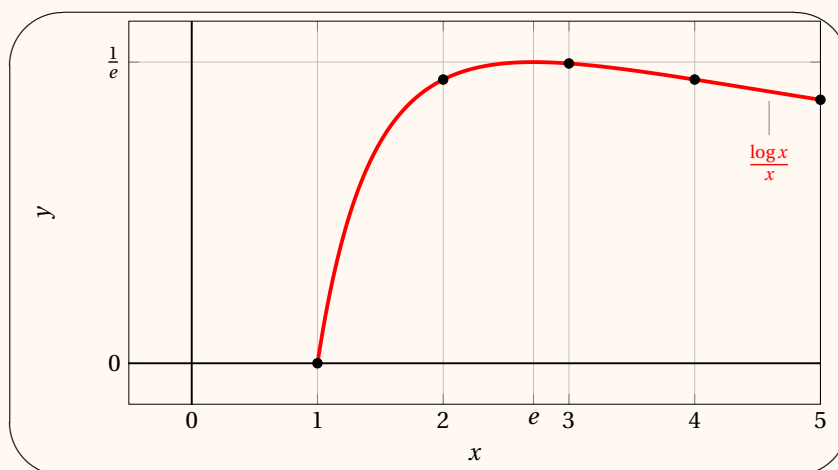
We pass from the discrete to the continuous and define the function

$$f(x) = \frac{\log x}{x}, \quad \forall x > 1$$

We have

$$f'(x) = \frac{1 - \log x}{x^2} \leq 0, \quad \forall x \geq e$$

and it follows that f is non-increasing $\forall x \geq e$. Consequently the sequence a_n is non-increasing for $n \geq 3$ (the first natural number $> e$).



- Be careful not to use the passage from the discrete to the continuous indiscriminately. For example, it cannot be done for the sequences $\frac{n!}{2^n}$ or $\frac{n^2 + (-1)^n n}{n^3 + 1}$, since they are defined only for natural numbers.